

1. main.c

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include "header.h"
```

```
int main(){
```

```
    int question;
```

```
    printf("Enter Question Numnber: ");
```

```
    scanf("%d", &question);
```

```
    getchar();
```

```
    switch(question){
```

```
        case 1:{
```

```
            float a, b;
```

```
            printf("Enter Two Numbers : ");
```

```
            scanf("%f%f", &a, &b);
```

```
            float max = MaxBetweenTwo(a , b);
```

```
            printf("Maximum Between %f and %f is : %f", a, b, max);
```

```
            break;
```

```
        }
```

```
        case 2:{
```

```
            float a;
```

```
            printf("Enter the Number : ");
```

```
            scanf("%f",&a);
```

```
            Check(a);
```

```
            break;
```

```
        }
```

```
        case 3:{
```

```
            int a;
```

```
            printf("Enter Number : ");
```

```
            scanf("%d", &a);
```

```
        CheckDivisibleBy5And11(a);
        break;
    }
case 4:{
    int a;

    printf("Enter Number : ");

    scanf("%d", &a);

    EvenOdd(a);

    break;
}
case 5:{
    int year;

    printf("Enter year : ");

    scanf("%d", &year);

    isLeapYear(year);

    break;
}
case 6:{
    char ch;

    printf("Enter a Character : ");

    scanf("%c", &ch);

    AlphabetCheck(ch);

    break;
}
case 7:{
    char ch;

    printf("Enter a Character : ");

    scanf("%c", &ch);

    VowelConsonant(ch);

    break;
}
```

```

case 8:{
    char ch;

    printf("Enter a Character : ");

    scanf("%c", &ch);

    AlphabetDigitSpecialChar(ch);

    break;
}

case 9:{
    char ch;

    printf("Enter a Character : ");

    scanf("%c", &ch);

    UppercaseLowercase(ch);

    break;
}

case 10:{
    int mn;

    printf("Enter Month Number : ");

    scanf("%d", &mn);

    int days = NumberOfDaysInMonth(mn);

    printf("Number of Days in month no %d are : %d", mn, days);

    break;
}

case 11:{
    float a , b , c;

    printf("Enter Angles of Triangle : ");

    scanf("%f%f%f", &a, &b, &c);

    AnglesOfTriangle(a, b, c);

    break;
}

case 12:{
    float a , b , c;

```

```

printf("Enter Angles of Triangle : ");
scanf("%f%f%f", &a, &b, &c);
TypeOfTriangle(a, b, c);
break;
}
case 13:{
    float a, b, c;
    printf("Enter Coefficients a, b, c of Quadratic Equation: ");
    scanf("%f%f%f", &a, &b, &c);
    RootsOfQuadratiEquation(a, b, c);
    break;
}
case 14:{
    float a, b, c, d, e;
    printf("Enter Marks of Five Subjects : ");
    scanf("%f%f%f%f%f", &a, &b, &c, &d, &e);
    char grade = Grade(a, b, c, d, e);
    printf("Your Grade is %c", grade);
    break;
}
case 15:{
    float basic_salary , gross_salary;
    printf("Enter Basic Salary : ");
    scanf("%f", &basic_salary);
    gross_salary = GrossSalary(basic_salary);
    printf("The gross salary of the employee is : %.2f\n", gross_salary);
    break;
}
case 16:{
    int units;
    float bill;

```

```
    printf("Enter Electricity Units : ");  
    scanf("%d", &units);  
    bill = ElectricityBill(units);  
    printf("Total Electricity Bill is %f ", bill);  
    break;  
}  
default:  
    break;  
}  
return 0;  
}
```

2. header.h

```
float MaxBetweenTwo(float a , float b);  
void Check(float a);  
void CheckDivisibleBy5And11(int a);  
void EvenOdd(int a);  
void isLeapYear(int year);  
void AlphabetCheck(char ch);  
void VowelConsonant(char ch);  
void AlphabetDigitSpecialChar(char ch);  
void UppercaseLowercase(char ch);  
int NumberOfDaysInMonth(int mn);  
void AnglesOfTriangle(float a, float b, float c);  
void TypeOfTriangle(float a, float b, float c);  
void RootsOfQuadratiEquation(float a, float b, float c);  
char Grade(float a , float b , float c, float d, float e);  
float GrossSalary(float basic_salary);  
float ElectricityBill(int units);
```

3. logic.c

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include "header.h"

#include <math.h>

#include <ctype.h>
```

```
float MaxBetweenTwo(float a , float b){

    if(a > b){

        return a;

    }

    else

        return b;

}
```

```
void Check(float a){

    if(a > 0){

        printf("The Number %f is Positive", a);

    }

    else if(a < 0){

        printf("The Number %f is Negative", a);

    }

    else{

        printf("The Number %f is Zero", a);

    }

}
```

```
void CheckDivisibleBy5And11(int a){

    if((a % 5 == 0) && (a % 11 == 0)){

        printf("The Number %d is divisible by both 5 and 11", a);

    }

}
```

```
}  
else{  
    printf("The Number %d is not divisible by both 5 and 11", a);  
}  
}
```

```
void EvenOdd(int a){  
    if(a % 2 == 0){  
        printf("The Number %d is Even",a);  
    }  
    else{  
        printf("The Number %d is Odd",a);  
    }  
}
```

```
void isLeapYear(int year){  
    if(year % 400 == 0){  
        printf("The Year %d is a Leap Year", year);  
    }  
    else if(year % 100 == 0){  
        printf("The Year %d is not aLeap Year", year);  
    }  
    else if(year % 4 == 0){  
        printf("The Year %d is a Leap Year", year);  
    }  
    else{  
        printf("The Year %d is not aLeap Year", year);  
    }  
}
```



```

void AlphabetCheck(char ch){
    int A = (int)ch;
    if((A >= 65 && A <= 90) || (A >= 97 && A <= 122)){
        printf("The Given Character %c is Alphabet", ch);
    }
    else{
        printf("The Given Character %c is not a Alphabet", ch);
    }
}

```

```

void VowelConsonant(char ch){
    char chr = tolower(ch);
    int a = (int)chr;
    if(a == 97 || a == 101 || a == 105 || a == 111 || a == 117){
        printf("The character %c is a vowel", ch);
    }
    else{
        printf("The character %c is a consonant", ch);
    }
}

```

```

void AlphabetDigitSpecialChar(char ch){
    int A = (int)ch;
    if((A >= 65 && A <= 90) || (A >= 97 && A <= 122)){
        printf("The Given Character %c is Alphabet", ch);
    }
    else if(A >= 48 && A <= 57){
        printf("The Given Character %c is Digit", ch);
    }
    else {
        printf("The Given Character %c is a Special Character", ch);
    }
}

```

```
}  
}
```

```
void UppercaseLowercase(char ch){  
    int A = (int)ch;  
    if(A >= 65 && A <= 90){  
        printf("The Character %c is Uppercase alphabet", ch);  
    }  
    else if(A >= 97 && A <= 122){  
        printf("The Character %c is Lowercase alphabet", ch);  
    }  
    else{  
        printf("The Character %c is not a alphabet", ch);  
    }  
}
```

```
int NumberOfDaysInMonth(int mn){  
    int NoOfDays = 0;  
    if(mn <= 0 || mn > 12){  
        printf("Invalid Month Number ");  
    }  
    else if(mn == 1 || mn == 3 || mn == 5 || mn == 7 || mn == 8 || mn == 10 || mn == 12){  
        NoOfDays = 31;  
    }  
    else if(mn == 2){  
        NoOfDays = 28;  
    }  
    else {  
        NoOfDays = 30;  
    }  
    return NoOfDays;  
}
```

```
}
```

```
void AnglesOfTriangle(float a, float b, float c){
```

```
    if(a + b + c == 180){
```

```
        printf("The Triangle is Valid");
```

```
    }
```

```
    else{
```

```
        printf("The Triangle is Invalid");
```

```
    }
```

```
}
```

```
void TypeOfTriangle(float a, float b, float c){
```

```
    if(a + b + c == 180){
```

```
        if(a == b && b == c && a == c){
```

```
            printf("The Triangle is Equilateral");
```

```
        }
```

```
        else if((a == b && a != c) || (b == c && a != b) || (a == c && b != c)){
```

```
            printf("The Triangle is Isosceles");
```

```
        }
```

```
        else{
```

```
            printf("The Triangle is Scalene");
```

```
        }
```

```
    }
```

```
    else{
```

```
        printf("The Triangle is Invalid");
```

```
    }
```

```
}
```

```
void RootsOfQuadratiEquation(float a, float b, float c){
```

```
    float root1, root2, realPart, imaginaryPart;
```

```
    float discriminant = b * b - 4 * a * c;
```

```

if(a == 0){
    printf("It is not a quadratic equation");
}
else if(discriminant > 0){
    root1 = (-b + sqrt(discriminant)) / (2 * a);
    root2 = (-b - sqrt(discriminant)) / (2 * a);
    printf("Roots of the quadratic equation are real and different\n");
    printf("Root1 = %f\n", root1);
    printf("Root2 = %f", root2);

}
else if(discriminant == 0){
    root1 = root2 = -b / (2 * a);
    printf("Roots of the quadratic equation are real and same\n");
    printf("Root1 = Root2 = %f\n",root1);
}
else{
    realPart = -b / (2 * a);
    imaginaryPart = sqrt(-discriminant) / (2 * a);
    printf("Roots of the quadratic equation are complex and different\n");
    printf("Root1 = %.2f + %.2fi\n", realPart, imaginaryPart);
    printf("Root2 = %.2f + %.2fi\n", realPart, imaginaryPart);
}
}

```

```

char Grade(float a , float b , float c, float d, float e){
    float percentage = (a + b + c + d + e)/5;
    printf("Your Percentage is %f\n", percentage);
    if(percentage >= 90){
        return 'A';
    }
}

```

```
else if(percentage >= 80){
    return 'B';
}
else if(percentage >= 70){
    return 'C';
}
else if(percentage >= 60){
    return 'D';
}
else if(percentage >= 40){
    return 'E';
}
else{
    return 'F';
}
}
```

```
float GrossSalary(float basic_salary){
    float gross_salary, hra, da;
    if(basic_salary <= 10000){
        hra = (0.20) * basic_salary;
        da = (0.80) * basic_salary;
    }
    else if(basic_salary <= 20000){
        hra = (0.25) * basic_salary;
        da = (0.90) * basic_salary;
    }
    else {
        hra = (0.30) * basic_salary;
        da = (0.95) * basic_salary;
    }
}
```

```
gross_salary = basic_salary + hra + da;  
return gross_salary;  
}
```

```
float ElectricityBill(int units){  
    float bill;  
    if(units <= 50){  
        bill = units * 0.50;  
    }  
    else if(units <= 150){  
        bill = 50 * 0.5 + (units - 50) * 0.75;  
    }  
    else if(units <= 250){  
        bill = 50 * 0.5 + 100 * 0.75 +(units - 150) * 1.20;  
    }  
    else if(units > 250){  
        bill = 50 * 0.5 + 100 * 0.75 + 100 * 1.20 + (units - 250) * 1.50;  
    }  
    bill = bill + bill * 20 / 100;  
    return bill;  
}
```

